

der哥 美股自动交易系统工程手册

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本文面向项目用户、维护者和策略开发者，目标是按软件工程方式说明系统边界、运行模式、文件职责、API/安全约束和专业术语。本文不是法律意见，也不是投资建议。

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1. 项目定义和权利说明

项	说明
项目名称	der哥，本手册中作为项目名称和版权声明中的权利标识使用。
法律身份	der哥 这个写法本身不自动等于公司、个人实名或注册商标；如果要做更强权利保护，应把 LICENSE 中的主体改成个人法定姓名或公司主体，并按目标司法辖区申请版权登记/商标注册。
授权方式	当前采用 proprietary / all rights reserved：未获书面许可，不得复制、转发、公开、修改、商用或再授权。
协作边界	给用户或协作者访问仓库，只代表授权阅读或协作，不代表开源许可。贡献代码的归属和使用权建议另签简单贡献协议。
投资边界	本项目是 IBKR paper trading 自动化实验系统，不是实盘投资建议系统。live trading 当前没有安全启用路径。

2. 当前系统状态和运行模式

当前菜单已经把常用入口收敛为主菜单 0/1/9/2：0 STOP 停止，1 DEV_LOCK 开发锁，9 MODE9_RUNTIME 自动确保 TRADE_LOCK API 并启动 autonomous agent，2 MORE_TOOLS 才展开高级/手动工具。

模式	当前位置	是否下单	核心含义
0 STOP	主菜单	否	停 API、停 agent、停 pool strategy、清 trade token。
1 DEV_LOCK	主菜单	否	开发/检查模式，kill switch 开，ALLOW_PAPER_TRANSMIT=false。
9 MODE9_RUNTIME	主菜单	可能	推荐自动化入口：确保 TRADE_LOCK API 后启动 autonomous agent；agent 按节奏监控股票池并调用策略。
2 MORE_TOOLS	主菜单	视子项	展开高级工具，不是交易锁状态。
2 TRADE_LOCK	高级菜单	不直接下单	启动 paper API、生成 daily token、允许 paper transmit 的前置状态。
3 HEALTH	高级菜单	否	查询 API/TWS/token/account readiness。
4 AUTO_VALIDATE_SEQ	高级菜单	否	生成结构化订单请求并调用 validate endpoint；不创建 TWS 订单。
5 AUTO_STAGE_SEQ	高级菜单	是，未传输	创建新的 TWS staged order，transmit=False，可能需要手动 Transmit 或取消；危险测试项。
6 AUTO_PAPER_SEQ	高级菜单	是，自动传输	创建新的 transmitted paper order，transmit=True；偏一次性测试。
7 POOL_STRATEGY_PAPER	高级菜单	可能	前台跑 pool strategy；日志显示在当前终端，策略语义和后台相同，若 mode=paper 会提交 paper endpoint。
8 POOL_STRATEGY_BG	高级菜单	可能	后台长期跑 pool strategy；差异是进程管理和日志位置，不是“只展示”。
10-22	高级菜单	多为报告/诊断	监控、市场会话、rollout precheck、full paper automation、discovery/news/simulation、account sync、callback dry-run 等。

常见误解澄清：validate 不是 TWS 虚拟单，它只是 Trading API 的校验请求；stage 才会进 TWS 但不 transmit；paper 会进 TWS paper 并自动 transmit。7 和 8 不是展示模式，它们都可能根据 POOL_STRATEGY_MODE 调用 validate/stage/paper；前台/后台只影响是否占用当前终端和日志在哪里看。

3. 架构总览

层	主要目录/文件	职责	禁止事项
控制入口	scripts/control_trading_mode. command, api_service.py	启停模式、暴露 HTTP API、统一入口。	不绕过 API 直连 broker。
数据层	data/, trading/market_data.py, trading/ibkr_readonly.py	股票池、security master、行情读取、bar/cache。	不用延迟/缺失行情做 fast execution。
研究层	openclaw/, trading/candidate_research.py, research/	AI/规则候选、报告、paper 证据链。	AI 不直接写 universe，不直接下单。
策略层	trading/strategy_modules.py, trading/strategy.py, scripts/run_pool_strategy_module.py	BUY/SELL/HOLD、组合风险、订单意图。	策略不保存 daily token，不连接 TWS 执行。
风控层	trading/risk.py, trading/session_policy.py, trading/event_risk_control.py	lock、token、allowlist、金额、风险、时段、事件风险。	不因行情缺失放松风险参数。
执行层	trading/service.py, trading/tws_paper.py, execution/	validate/stage/paper、TWS paper order、审计。	live trading 当前禁用。
审计/展示	trading/audit.py, reports/, dashboard/	SQLite audit、JSON/MD/HTML 报告、dashboard。	不手改 trading_audit.sqlite3。

4. 关键业务语义

主题	工程解释
MODE9_RUNTIME	当前推荐自动化入口；它会确保 TRADE_LOCK API 后启动 autonomous agent。agent 周期性做 health、universe、quote、risk、strategy、report。
POOL_STRATEGY_PAPER/BG	同一个 pool strategy runner 的前台/后台运行方式；前台便于观察，后台适合长跑。只要 mode=paper 且风控通过，就可能提交 paper order。
AUTO_VALIDATE_SEQ	自动构造测试/候选 payload 并请求 API 校验；不会提交 TWS，不会生成 staged order。
AUTO_STAGE_SEQ	自动构造 payload 并创建 TWS untransmitted order；这不是安全展示，可能在 TWS 留下待手动处理订单。
AUTO_PAPER_SEQ	自动构造 payload 并提交 transmitted paper order；需要 TRADE_LOCK、token、TWS ready 和风控通过。
bracket order	parent limit entry + child stop；parent 通常 transmit=False，child stop transmit 时整组提交。不要简单把 parent 改为 transmit=True。
single limit order	pool strategy 当前主要提交 single limit order；是否 transmit 由 endpoint mode 决定。
audit ledger	SQLite 本地审计，不是交易路径的一部分；用于证明请求、失败、TWS acknowledgement 和耗时。
generated artifacts	reports/, dashboard/, logs/, .runtime/ 是运行输出；手册按目录说明，不逐个产物维护。

5. 用户上手示例

场景	操作	预期结果
只读代码	读 README.md、本手册、docs/PROJECT_MAP.md、docs/STRATEGY_MODULES.md。	了解边界和主流程，不触碰 TWS。
本地模拟策略	.venv313/bin/python scripts/run_strategy_simulation.py --symbols AAPL,MSFT --steps 12	不连接 IBKR，不下单，只验证 quote/cache/bar/signal。
validate 策略	.venv313/bin/python scripts/run_pool_strategy_module.py --mode validate --source simulated-scenario --symbols AAPL,MSFT,AMD,INTC,KO,PFE,T,F --steps 18 --batch-size 4	调 API validate 或本地模拟路径，不创建 TWS order。
查看运行状态	打开 reports/autonomous_agent/latest.json、logs/autonomous_agent.log。	判断 agent 是否运行、行情是否缺失、是否有订单提交。
只读 IBKR 行情	.venv313/bin/python scripts/check_ibkr_quotes.py --symbols AAPL,MSFT,QQQ --market-data-type 3	检查 delayed/read-only quote，不提交订单。
推荐自动 paper 入口	运行 scripts/control_trading_mode.command，选择 9 MODE9_RUNTIME。	自动确保 TRADE_LOCK API 并启动 agent；仍受所有风控限制。

6. 工程目录与逐文件索引

说明：本节以目录为大类列出 git 已跟踪源文件。作用 来自模块 docstring、一级标题或文件命名；关键函数/类 通过 AST/脚本函数名抽取，描述保持短句，便于快速定位。

根目录

文件	作用	关键函数/类
.env.example	环境变量示例，不能放真实 secret	无函数；作为文档、配置或数据输入使用
.gitignore	Git 忽略规则	无函数；作为文档、配置或数据输入使用
.pre-commit-config.yaml	pre-commit 与 gitleaks 配置	无函数；作为文档、配置或数据输入使用
AGENTS.md	Repository Instructions	无函数；作为文档、配置或数据输入使用
ARCHITECTURE.md	Architecture	无函数；作为文档、配置或数据输入使用
AUDIT_MODEL.md	Audit Model	无函数；作为文档、配置或数据输入使用
DATA_FEEDS.md	Data Feeds	无函数；作为文档、配置或数据输入使用
LICENSE	专有版权与使用限制条款	无函数；作为文档、配置或数据输入使用
README.md	IBKR Paper Trading Gateway	无函数；作为文档、配置或数据输入使用
SECURITY_MODEL.md	Security Model	无函数；作为文档、配置或数据输入使用
TRADING_LOCK.md	Trading Lock Policy	无函数；作为文档、配置或数据输入使用
VALUE_POOL.md	Value Pool	无函数；作为文档、配置或数据输入使用
api_service.py	Authenticated JSON API for OpenClaw-to-Python trading requests.	类 TradingApiHandler(do_GET, do_POST, log_message, _authenticated, _read_json, _send_json); main:命令入口
main.py	项目文件	main:命令入口
run_dry_run_api.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
twse_readonly_check.py	Verify a read-only connection to an IBKR paper-trading TWS session.	类 ReadOnlyTwsClient(__init__, nextValidId, currentTime, managedAccounts, accountSummary, accountSummaryEnd, error); parse_args:解析命令行参数; main:命令入口

.codex

文件	作用	关键函数/类
.codex/hooks.json	JSON 配置或钩子定义	无函数；作为文档、配置或数据输入使用

config

文件	作用	关键函数/类
config/__init__.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
config/autonomous_paper.env.example	环境变量示例，不能放真实 secret	无函数；作为文档、配置或数据输入使用
config/defaults.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化

data

文件	作用	关键函数/类
data/security_master.csv	项目数据表/股票池输入	无函数；作为文档、配置或数据输入使用
data/us_equity_universe.csv	项目数据表/股票池输入	无函数；作为文档、配置或数据输入使用

docs

文件	作用	关键函数/类
docs/BOOK_BASED_RESEARCH_ROADMAP.md	Book-Based Quant Research Roadmap	无函数；作为文档、配置或数据输入使用
docs/HARDWARE_SOFTWARE_ROADMAP.md	Hardware And Software Roadmap	无函数；作为文档、配置或数据输入使用
docs/NEXT_PHASE_AUTONOMOUS_PAPER_TRADING_PROMPT.md	Next Phase Prompt: Autonomous IBKR Paper Trading Runtime	无函数；作为文档、配置或数据输入使用
docs/PROJECT_HANDOFF_MANUAL_CN.md	der哥 美股自动交易系统工程手册	无函数；作为文档、配置或数据输入使用
docs/PROJECT_MAP.md	Project Map	无函数；作为文档、配置或数据输入使用
docs/RESEARCH_CANDIDATE_PIPELINE.md	Research Candidate Pipeline	无函数；作为文档、配置或数据输入使用

文件	作用	关键函数/类
docs/STRATEGY_DEBT.md	Strategy Debt	无函数；作为文档、配置或数据输入使用
docs/STRATEGY_FLOW.md	Strategy Flow	无函数；作为文档、配置或数据输入使用
docs/STRATEGY_MODULES.md	OpenClaw Strategy Modules	无函数；作为文档、配置或数据输入使用
docs/YUGE_AUTONOMOUS_TRADING_BRIEF.md	OpenClaw / IBKR Paper 自动投资系统说明	无函数；作为文档、配置或数据输入使用

docs/trading_sessions

文件	作用	关键函数/类
docs/trading_sessions/01_calendar_and_timezones.md	Calendar And Timezones	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/02_overnight_session.md	Overnight Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/03_premarket_session.md	Premarket Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/04_regular_session.md	Regular Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/05_afterhours_session.md	Afterhours Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/06_risk_by_session.md	Risk By Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/07_order_types_by_session.md	Order Types By Session	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/08_monitoring_and_recovery.md	Monitoring And Recovery	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/09_autonomous_runtime_runbook.md	Autonomous Runtime Runbook	无函数；作为文档、配置或数据输入使用
docs/trading_sessions/README.md	Trading Sessions	无函数；作为文档、配置或数据输入使用

execution

文件	作用	关键函数/类
execution/__init__.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
execution/paper.py	项目文件	类 PaperOrder; paper_order:模块辅助逻辑; live_order:模块辅助逻辑

openclaw

文件	作用	关键函数/类
openclaw/SKILL.md	IBKR Paper Gateway	无函数；作为文档、配置或数据输入使用
openclaw/openclaw_trading_client.py	OpenClaw client for the Mac Python IBKR paper-trading gateway.	attach_workflow_step:模块辅助逻辑; workflow_error_payload:模块辅助逻辑; load_secret:读取/加载数据; request_json:发起请求; parse_args:解析命令行参数; add_order_args:模块辅助逻辑; order_payload:模块辅助逻辑; preflight_paper_order:模块辅助逻辑; exit_code_for_result:模块辅助逻辑; add_client_timing:模块辅助逻辑; add_execution_mode:模块辅助逻辑; require_trade_session_token:模块辅助逻辑; _elapsed_ms:内部辅助; _utc_now:内部辅助; main:命令入口

ops/launchd

文件	作用	关键函数/类
ops/launchd/com.openclaw.autonomous-paper.plist	macOS launchd 后台任务配置	定义后台启动命令、环境变量和进程标签
ops/launchd/com.openclaw.mode9-agent.plist	macOS launchd 后台任务配置	定义后台启动命令、环境变量和进程标签

research

文件	作用	关键函数/类
research/gitkeep	项目文件	无函数
research/__init__.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
research/full_paper_trial_pipeline.py	项目文件	类 FullPaperTrialPipelineConfig; 类 FullPaperTrialPipelineReport; run_full_pipeline:运行流程; main:命令入口; _overall_status:内部辅助; _artifacts:内部辅助
research/guarded_paper_trial_session.py	项目文件	类 GuardedPaperTrialSessionConfig; 类 GuardedPaperTrialSessionReport; run_guarded_session:运行流程; main:命令入口; _load_or_fetch_health:内部辅助; _session_report:内部辅助; _session_status:内部辅助; _artifacts:内部辅助; _write_session_report:内部辅助
research/ibkr_mean_reversion_pipeline.py	项目文件	类 IbkrMeanReversionPipelineConfig; 类 IbkrMeanReversionPipelineReport; run_pipeline:运行流程; main:命令入口; _diagnose_csv:内部辅助; _paper_plan:内部辅助; _pipeline_report:内部辅助
research/ibkr_quote_recorder.py	项目文件	类 QuoteSource(get_quotes); 类 QuoteRecorderConfig; 类 QuoteRecorderReport; record_quotes:记录数据; main:命令入口; _quote_row:内部辅助; _new_errors:内部辅助
research/mean_reversion_diagnostics.py	项目文件	类 MeanReversionDiagnostics; 类 PriceDiagnostics; diagnose_prices:模块辅助逻辑; diagnose_symbol_prices:模块辅助逻辑; read_prices_csv:读取输入; write_report:写出文件或结果; main:命令入口; _lag1_adf_t_stat:内部辅助; _half_life:内部辅助; _hurst_exponent:内部辅助; _variance_ratio:内部辅助; _latest_z_score:内部辅助; _ols_slope:内部辅助; _slope_t_stat:内部辅助; _sample_variance:内部辅助; _round_optional:内部辅助
research/paper_candidate_hunt.py	项目文件	类 CandidateHuntAttempt; 类 PaperCandidateHuntReport; 类 PaperCandidateHuntConfig; run_candidate_hunt:运行流程; main:命令入口; _candidate_found:内部辅助; _hunt_status:内部辅助; _int_value:内部辅助; _load_optional_json:内部辅助

文件	作用	关键函数/类
research/paper_execution_gate.py	项目文件	类 PaperExecutionGateReport; load_json_object:读取/加载数据; build_gate_report:构建对象/报告; execute_if_confirmed:模块辅助逻辑; write_report:写出文件或结果; main:命令入口; _selected_candidate:内部辅助; _health_block_reason:内部辅助; _pre_submit_review_block_reason:内部辅助; _load_optional_json:内部辅助; _replace_report:内部辅助; _request_json:内部辅助
research/paper_pre_submit_review.py	项目文件	类 PreSubmitReviewItem; 类 PaperPreSubmitReview; build_pre_submit_review:构建对象/报告; write_review_json:写出文件或结果; write_review_markdown:写出文件或结果; main:命令入口; _item:内部辅助; _one_share_buy_payload:内部辅助; _payload_detail:内部辅助; _pnl_detail:内部辅助; _status:内部辅助; _value:内部辅助; _mapping_or_none:内部辅助; _load_optional_json:内部辅助; _load_last_jsonl:内部辅助
research/paper_readiness_report.py	项目文件	类 ReadinessItem; 类 PaperReadinessReport; load_plan:读取/加载数据; build_readiness_report:构建对象/报告; write_readiness_report:写出文件或结果; main:命令入口; _readiness_items:内部辅助; _operator_steps:内部辅助
research/paper_session_runbook.py	项目文件	类 RunbookCommand; 类 PaperSessionRunbook; build_runbook:构建对象/报告; write_runbook_json:写出文件或结果; write_runbook_markdown:写出文件或结果; render_markdown:渲染输出; main:命令入口; _blockers:内部辅助; _runbook_status:内部辅助; _summary:内部辅助; _operator_steps:内部辅助; _commands:内部辅助; _artifacts:内部辅助; _load_json_object:内部辅助; _load_optional_json_object:内部辅助; _dedupe:内部辅助
research/paper_session_status_page.py	项目文件	render_status_page:渲染输出; write_status_page:写出文件或结果; main:命令入口; _state_panel:内部辅助; _list_panel:内部辅助; _ordered_panel:内部辅助; _commands_panel:内部辅助; _artifacts_panel:内部辅助; _status_class:内部辅助; _css:内部辅助; _load_json_object:内部辅助
research/paper_trial_evidence_bundle.py	项目文件	类 EvidenceArtifact; 类 PaperTrialEvidenceBundle; build_evidence_bundle:构建对象/报告; write_evidence_bundle:写出文件或结果; write_evidence_markdown:写出文件或结果; main:命令入口; _stage:内部辅助; _summary:内部辅助; _next_actions:内部辅助; _artifacts:内部辅助; _artifact_status:内部辅助; _artifact_created_at:内部辅助; _status:内部辅助; _load_optional_json:内部辅助
research/paper_trial_preflight_checklist.py	项目文件	类 PreflightCheck; 类 PaperTrialPreflightChecklist; build_preflight_checklist:构建对象/报告; write_preflight_json:写出文件或结果; write_preflight_markdown:写出文件或结果; render_markdown:渲染输出; main:命令入口; _machine_checks:内部辅助; _status:内部辅助; _next_allowed_action:内部辅助; _manual_confirmations:内部辅助; _stop_conditions:内部辅助; _commands:内部辅助; _rehearsal_is_clean:内部辅助; _rehearsal_detail:内部辅助; _load_json_object:内部辅助; _load_optional_json_object:内部辅助
research/paper_trial_readiness_monitor.py	项目文件	类 ReadinessMonitorSnapshot; 类 PaperTrialReadinessMonitorReport; 类 PaperTrialReadinessMonitorConfig; run_readiness_monitor:运行流程; main:命令入口; _monitor_status:内部辅助; _load_optional_json_object:内部辅助
research/paper_trial_reconciliation.py	项目文件	类 ReconciliationItem; 类 PaperTrialReconciliationReport; build_reconciliation:构建对象/报告; write_reconciliation:写出文件或结果; main:命令入口; _audit_item:内部辅助; _twos_order_item:内部辅助; _pnl_item:内部辅助; _twos_snapshot_contains_key:内部辅助; _report:内部辅助; _idempotency_from_gate:内部辅助; _load_audit_order:内部辅助; _load_optional_json_object:内部辅助; _load_latest_jsonl:内部辅助; _optional_str:内部辅助
research/paper_trial_rehearsal.py	项目文件	类 PaperTrialRehearsalReport; 类 RehearsalQuoteSource(_init_, get_quotes); run_rehearsal:运行流程; main:命令入口; _rehearsal_settings:内部辅助
research/paper_trial_report_refresh.py	项目文件	类 PaperTrialReportRefreshConfig; 类 PaperTrialReportRefreshReport; refresh_reports:刷新状态; main:命令入口; 类 _BlockedQuoteSource(get_quotes); _artifacts:内部辅助; _load_optional_json:内部辅助; _load_latest_jsonl:内部辅助; _idempotency_from_gate:内部辅助; _load_audit_order:内部辅助
research/paper_validation_plan.py	项目文件	类 PaperValidationCandidate; 类 PaperValidationPlan; build_plan:构建对象/报告; load_diagnostics:读取/加载数据; write_plan:写出文件或结果; submit_validate_payloads:提交请求; main:命令入口; _candidate_from_item:内部辅助; _validation_payload:内部辅助; _optional_float:内部辅助; _request_json:内部辅助
research/project_completion_audit.py	项目文件	类 CompletionRequirement; 类 ProjectCompletionAudit; build_completion_audit:构建对象/报告; write_completion_audit_json:写出文件或结果; write_completion_audit_markdown:写出文件或结果; render_markdown:渲染输出; main:命令入口; _requirement:内部辅助; _summary:内部辅助; _next_required_evidence:内部辅助; _clean_rehearsal:内部辅助; _rehearsal_detail:内部辅助; _stage:内部辅助; _quote_capture_done:内部辅助; _validate_done:内部辅助; _validate_detail:内部辅助; _status:内部辅助; _value:内部辅助; _load_optional_json:内部辅助
research/quote_quality_report.py	项目文件	类 SymbolQuoteQuality; 类 QuoteQualityReport; build_quote_quality_report:构建对象/报告; write_quote_quality_json:写出文件或结果; write_quote_quality_markdown:写出文件或结果; render_markdown:渲染输出; main:命令入口; _symbol_quality:内部辅助; _status:内部辅助; _next_actions:内部辅助; _read_rows:内部辅助; _float:内部辅助; _format_optional:内部辅助
research/strategy_catalog.py	项目文件	类 StrategyCatalogEntry; 类 StrategyCatalog; build_strategy_catalog:构建对象/报告; write_strategy_catalog_json:写出文件或结果; write_strategy_catalog_markdown:写出文件或结果; render_markdown:渲染输出; main:命令入口
research/strategy_validation_plan.py	项目文件	类 StrategyValidationPlanConfig; build_strategy_validation_plan:构建对象/报告; main:命令入口; _load_quote_rows:内部辅助; _group_rows:内部辅助; _market_data_from_rows:内部辅助; _strategy_signals:内部辅助; _candidate_from_signal:内部辅助; _validation_payload:内部辅助; _row_price:内部辅助; _spread_bps:内部辅助; _parse_timestamp:内部辅助; _optional_float:内部辅助

risk

文件	作用	关键函数/类
risk/_init__.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
risk/core.py	项目文件	类 OrderRiskConfig; 类 OrderRiskDecision; validate_order:校验规则

scripts

文件	作用	关键函数/类
scripts/approve_candidate_to_universe.py	Approve a hard-audited candidate report into the local universe CSV.	parse_args:解析命令行参数; main:命令入口; read_rows:读取输入; write_rows:写出文件或结果; upsert_candidate:模块辅助逻辑; row_from_report:模块辅助逻辑
scripts/audit_ibkr_level1_streaming.py	Read-only IBKR Level 1 streaming capability audit.	类 Level1AuditClient(__init__, nextValidId, contractDetails, contractDetailsEnd, marketDataType, tickPrice, tickSize, error, run_loop, _row); main:命令入口; _stock_contract:内部辅助
scripts/audit_paper_trading_readiness.py	Audit local readiness for a one-share IBKR paper trial.	无顶层类/函数，主要为常量、导入或包初始化
scripts/auto_paper_limit_from_quote.py	Create a paper limit order from the latest IBKR quote.	parse_args:解析命令行参数; main:命令入口; latest_quote:模块辅助逻辑; order_payload:模块辅助逻辑; calculate_limit_price:模块辅助逻辑; quote_spread_bps:模块辅助逻辑; quote_payload:模块辅助逻辑; request_json:发起请求; read_required:读取输入; read_trade_session_token:读取输入; step:模块辅助逻辑; elapsed_ms:模块辅助逻辑
scripts/benchmark_autonomous_agent_cycle.py	Benchmark autonomous-agent cycle timings across universe sizes.	parse_args:解析命令行参数; main:命令入口; summarize_size:汇总信息
scripts/benchmark_ibkr_quote_batches.py	Benchmark read-only IBKR quote request latency by batch size.	parse_args:解析命令行参数; main:命令入口; parse_symbols:模块辅助逻辑; parse_batch_sizes:模块辅助逻辑; summarize:汇总信息
scripts/build_account_reports.py	Build read-only account, position, open-order, and daily summary reports.	main:命令入口; latest_pnl_row:模块辅助逻辑; flatten_open_orders:模块辅助逻辑; daily_summary:模块辅助逻辑; audit_requests_for_date:审计状态; write_json:写出文件或结果; float_or_none:模块辅助逻辑
scripts/build_ibkr_callback_wiring_audit.py	Build IBKR callback wiring audit reports.	main:命令入口
scripts/build_market_data_audit_report.py	项目文件	main:命令入口; _read_json:内部辅助; _latest_pool_report:内部辅助; _code_hits:内部辅助; _execution_uses_delayed:内部辅助; _runtime_market_data_type:内部辅助; _symbols_with_valid_bid_ask:内部辅助; _symbols_with_stale_quotes:内部辅助; _symbols_with_no_permission:内部辅助
scripts/build_market_data_billing_safety.py	Audit market data billing safety for IBKR reqMktData calls.	main:命令入口; build_report:构建对象/报告; audit_file:审计状态; enclosing_function:模块辅助逻辑; arg_value:模块辅助逻辑; literal:模块辅助逻辑; symbol_source:模块辅助逻辑; market_data_type_nearby:模块辅助逻辑; action_taken:模块辅助逻辑; env_bool:模块辅助逻辑; write_report:写出文件或结果
scripts/build_market_data_line_report.py	项目文件	main:命令入口; _read_json:内部辅助
scripts/build_mode9_execution_audit.py	项目文件	main:命令入口; _read_json:内部辅助; _read_text:内部辅助; _bool_env:内部辅助
scripts/build_mode9_infrastructure_status.py	Build unified Mode 9 infrastructure status and audit reports.	main:命令入口; build_all_reports:构建对象/报告; seed_current_sqlite_snapshots:模块辅助逻辑; ensure_sqlite_tables:确保前置条件; build_sqlite_audit:构建对象/报告; build_simple_table_report:构建对象/报告; build_lot_protection_report:构建对象/报告; build_outside_rth_protection_report:构建对象/报告; build_fixed_symbol_audit:构建对象/报告; build_cycle_audit:构建对象/报告; build_infrastructure_status:构建对象/报告; feature_status:模块辅助逻辑; build_final_audit:构建对象/报告; table_exists:模块辅助逻辑; read_json:读取输入; write_json_report:写出文件或结果; write_extra_report:写出文件或结果; generated_report_paths:生成内容; env_bool:模块辅助逻辑; feature_enabled:模块辅助逻辑; execution_feature:模块辅助逻辑; implementation_state:模块辅助逻辑; execution_state:模块辅助逻辑; category_for:模块辅助逻辑; report_mtime:模块辅助逻辑; last_action:模块辅助逻辑; remaining_gap:模块辅助逻辑; next_step:模块辅助逻辑; lot_protection_md:模块辅助逻辑; outside_rth_md:模块辅助逻辑; fixed_symbol_md:模块辅助逻辑; cycle_md:模块辅助逻辑; `infr
scripts/build_paper_execution_gate.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_pre_submit_review.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_readiness_report.py	Build a manual paper readiness report from a validate-only plan.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_session_runbook.py	Generate a paper-trial operator runbook from current reports.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_session_status_page.py	Generate a static HTML paper-trial status page.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_trial_evidence_bundle.py	Build a current-state evidence bundle for the paper-trial workflow.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_trial_preflight_checklist.py	Build a one-share IBKR paper-trial preflight checklist.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_trial_reconciliation.py	Build a post-paper-trial reconciliation report.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_paper_validation_plan.py	Build a validate-only paper trading plan from diagnostics JSON.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_pool_architecture_audit.py	Audit whether the six-layer pool architecture is implemented and wired.	main:命令入口; build_reports:构建对象/报告; layer_report:模块辅助逻辑; membership_records:模块辅助逻辑; record:记录数据; write_outputs:写出文件或结果; markdown:模块辅助逻辑; read_universe:读取输入; read_json:读取输入; sqlite_tables:模块辅助逻辑; membership_schema_complete:模块辅助逻辑
scripts/build_position_guard_report.py	项目文件	main:命令入口
scripts/build_project_completion_audit.py	Build the current project completion audit.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_quote_quality_report.py	Build a quote quality report from recorded IBKR quote samples.	无顶层类/函数，主要为常量、导入或包初始化
scripts/build_security_master.py	Build the persistent global security master.	main:命令入口; load_nasdaq_trader_url:读取/加载数据; load_nasdaq_trader_file:读取/加载数据; parse_pipe_text:模块辅助逻辑; write_records:写出文件或结果
scripts/build_strategy_catalog.py	Build the strategy module catalog.	无顶层类/函数，主要为常量、导入或包初始化

文件	作用	关键函数/类
scripts/build_strategy_extension_status.py	Build report-only status for strategy extensions not allowed to trade yet.	build_status:构建对象/报告; latest_agent_report:模块辅助逻辑; write_reports: 写出文件或结果; _bool_env:内部辅助; main:命令入口
scripts/build_strategy_validation_plan.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
scripts/check_gateway_health.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/check_ibkr_quotes.py	Fetch read-only IBKR quotes and print the normalized cache snapshot.	parse_args:解析命令行参数; main:命令入口
scripts/configure_feishu_webhook.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/control_trading_mode.command	macOS 可双击/终端运行脚本	vm_ssh_ready:模块辅助逻辑; write_trade_token_file:写出文件或结果; clear_trade_token_file:清理状态; stop_api_if_running:停止进程/服务; stop_pool_strategy_if_running:停止进程/服务; stop_autonomous_agent_if_running:停止进程/服务; default_universe_symbols:模块辅助逻辑; api_health_ok:模块辅助逻辑; api_health_field:模块辅助逻辑; detect_api_url:模块辅助逻辑; detect_trade_lock_api_url:模块辅助逻辑; load_trade_token_env:读取/加载数据; run_gateway_health:运行流程; run_auto_sequence:运行流程; run_pool_strategy_module:运行流程; start_pool_strategy_background:启动进程/服务; start_autonomous_agent_background:启动进程/服务; set_dev_lock_env:模块辅助逻辑; set_trade_lock_env:模块辅助逻辑; set_mode9_runtime_env:模块辅助逻辑; start_api_background:启动进程/服务; ensure_trade_lock_api_background:确保前置条件; run_monitor_on:运行流程; run_monitor_off:运行流程; run_monitor_status:运行流程; set_safe_report_env:模块辅助逻辑; run_market_session:运行流程; run_rollout_precheck:运行流程; `run_full_paper_autom
scripts/deploy_openclaw_gateway.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/diagnose_market_data_channels.py	Diagnose read-only IBKR market data channels in order.	parse_args:解析命令行参数; main:命令入口
scripts/generate_daily_feishu_update.py	Generate a sanitized daily OpenClaw project update for Feishu.	parse_args:解析命令行参数; read_json:读取输入; files_modified_since:模块辅助逻辑; universe_summary:模块辅助逻辑; latest_pnl:模块辅助逻辑; parse_timestamp:模块辅助逻辑; api_running:模块辅助逻辑; api_health_status:模块辅助逻辑; file_age_seconds:模块辅助逻辑; top_skip_reasons:模块辅助逻辑; explain_zero_submissions:模块辅助逻辑; build_status_snapshot:构建对象/报告; status_lines:模块辅助逻辑; api_running_legacy:模块辅助逻辑; observed_date:模块辅助逻辑; nth_weekday:模块辅助逻辑; last_weekday:模块辅助逻辑; easter_date:模块辅助逻辑; market_holidays:模块辅助逻辑; next_market_day:模块辅助逻辑; market_note:模块辅助逻辑; fmt_number:模块辅助逻辑; mask_identifier:模块辅助逻辑; sector_lines:模块辅助逻辑; pnl_lines:模块辅助逻辑; build_report:构建对象/报告; main:命令入口
scripts/generate_strategy_dashboard.py	Generate a local static strategy dashboard.	parse_args:解析命令行参数; main:命令入口
scripts/hunt_paper_candidate.py	Hunt for a validate payload without paper order submission.	无顶层类/函数，主要为常量、导入或包初始化
scripts/list_tws_orders.py	List current TWS open orders and recent executions.	类 ReadOnlyOrdersClient(__init__, nextValidId, error, openOrder, openOrderEnd, completedOrder, completedOrdersEnd, orderStatus, execDetails, execDetailsEnd, +1); _number:内部辅助; parse_args:解析命令行参数; main:命令入口; read_tws_orders:读取输入
scripts/monitor_paper_trial_readiness.py	Safely monitor readiness for a one-share IBKR paper trial.	无顶层类/函数，主要为常量、导入或包初始化
scripts/record_account_pnl.py	Record read-only IBKR paper account P&L samples over time.	类 PnlSample(as_dict); 类 AccountPnlClient(__init__, nextValidId, managedAccounts, accountSummary, accountSummaryEnd, pnl, updatePortfolio, error, run_loop, sample); parse_args:解析命令行参数; main:命令入口; append_outputs:追加记录; _float_or_none:内部辅助; _first_not_none:内部辅助; _sum_available:内部辅助
scripts/record_ibkr_quotes.py	Record read-only IBKR quote snapshots to CSV for research.	无顶层类/函数，主要为常量、导入或包初始化
scripts/refresh_paper_trial_reports.py	Refresh all paper-trial reports in safe order.	无顶层类/函数，主要为常量、导入或包初始化
scripts/refresh_position_protection_test_result.py	Refresh position protection test result from read-only TWS open-order output.	refresh_result:刷新状态; _find_order:内部辅助; main:命令入口
scripts/refresh_strategy_dashboard.py	Regenerate the static strategy dashboard on a timer.	parse_args:解析命令行参数; main:命令入口
scripts/rehearse_paper_trial_workflow.py	Run a simulated ready-environment rehearsal of the paper-trial workflow.	无顶层类/函数，主要为常量、导入或包初始化
scripts/review_candidate_report.py	Build a Python hard-audit report from an OpenClaw/Gemini candidate draft.	parse_args:解析命令行参数; main:命令入口; fetch_quote:拉取数据
scripts/review_t_duplicate_stops.py	项目文件	main:命令入口
scripts/run_auto_open_discovery_news_simulation.py	Run auto-open discovery/news/dynamic-pool/local-simulation pipeline.	main:命令入口
scripts/run_auto_order_sequence.py	Run an automatic module-style order sequence from current IBKR quotes.	parse_args:解析命令行参数; main:命令入口; fetch_quotes:拉取数据; require_trade_lock:模块辅助逻辑; evaluate_symbol:评估决策; calculate_limit_price:模块辅助逻辑; quote_spread_bps:模块辅助逻辑; quote_payload:模块辅助逻辑; endpoint_for_mode:模块辅助逻辑; request_json:发起请求; parse_symbols:模块辅助逻辑; read_required:读取输入; read_trade_session_token:读取输入; write_report:写出文件或结果; utc_now:模块辅助逻辑; elapsed_ms:模块辅助逻辑
scripts/run_autonomous_paper_runtime.py	项目文件	main:命令入口; _replace:内部辅助

文件	作用	关键函数/类
scripts/run_autonomous_trading_agent.py	Run the autonomous paper trading supervisor loop.	类 AgentCycle(as_dict); parse_args:解析命令行参数; main:命令入口; run_cycle:运行流程; failed_cycle:模块辅助逻辑; build_quote_readiness:构建对象/报告; evaluate_lifecycle_state:评估决策; lifecycle:模块辅助逻辑; sync_canonical_pool_state:模块辅助逻辑; pool_manager_records:模块辅助逻辑; quote_payloads_by_symbol:模块辅助逻辑; should_run_strategy_now:模块辅助逻辑; next_check_at:模块辅助逻辑; cycle_markdown:模块辅助逻辑; build_streaming_source:构建对象/报告; streaming_report_payload:模块辅助逻辑; select_monitor_symbols:筛选对象; fetch_quotes:拉取数据; update_market_state:更新状态; build_research_tasks:构建对象/报告; process_candidate_inbox:模块辅助逻辑; run_strategy_once:运行流程; mode9_buy_freeze:模块辅助逻辑; env_bool:模块辅助逻辑; request_json:发起请求; read_required:读取输入; append_jsonl:追加记录; write_json:写出文件或结果; write_text:写出文件或结果; utc_now:模块辅助逻辑; elapsed_ms:模块辅助逻辑
scripts/run_full_paper_automation.py	Run staged Mode 9 full paper automation enablement.	main:命令入口; run_once:运行流程; configure_full_paper_env:模块辅助逻辑; build_preflight:构建对象/报告; build_trade_pool_decisions:构建对象/报告; build_paper_buy_report:构建对象/报告; build_options_execution_report:构建对象/报告; build_gap_escape_execution_result:构建对象/报告; build_full_paper_run_report:构建对象/报告; build_final_audit:构建对象/报告; write_full_report:写出文件或结果; write_final_audit:写出文件或结果; insert_cycle:模块辅助逻辑; insert_market_session_event:模块辅助逻辑; markdown_full_report:模块辅助逻辑; markdown_final_audit:模块辅助逻辑; env_bool:模块辅助逻辑; now:模块辅助逻辑
scripts/run_full_paper_rollout.py	Run gated PAPER-only Mode 9 rollout stages.	main:命令入口; configure_safe_env:模块辅助逻辑; stage0_preflight:模块辅助逻辑; stage1_readiness:模块辅助逻辑; current_position_and_open_order_symbols:模块辅助逻辑; market_session_blocks_quote_wait:模块辅助逻辑; stage2_protective_repair:模块辅助逻辑; run_record_account_pnl_once:运行流程; stage2_precheck:模块辅助逻辑; stage2_blocked:模块辅助逻辑; build_stage2_result:构建对象/报告; protection_after_readback:模块辅助逻辑; write_position_protection_execution_result:写出文件或结果; write_stage:写出文件或结果; write_summary:写出文件或结果; stage_md:模块辅助逻辑; failed_checks:模块辅助逻辑; read_json:读取输入; env_bool:模块辅助逻辑; now:模块辅助逻辑
scripts/run_full_paper_trial_pipeline.py	Run the full read-only-to-paper-gate trial pipeline.	无顶层类/函数，主要为常量、导入或包初始化
scripts/run_guarded_paper_trial_session.py	Run the guarded IBKR paper-trial session workflow.	无顶层类/函数，主要为常量、导入或包初始化
scripts/run_ibkr_callback_dry_run.py	Run read-only REAL_IBKR callback bridge dry-run.	main:命令入口
scripts/run_ibkr_mean_reversion_pipeline.py	Record read-only IBKR quotes and run mean-reversion diagnostics.	无顶层类/函数，主要为常量、导入或包初始化
scripts/run_mean_reversion_diagnostics.py	Run lightweight mean-reversion diagnostics on a CSV price file.	无顶层类/函数，主要为常量、导入或包初始化
scripts/run_mode9_report_only_infrastructure_cycle.py	Refresh Mode 9 report-only infrastructure without submitting orders.	main:命令入口; run_report_only_cycle:运行流程; capture:模块辅助逻辑; summarize:汇总信息; mark_state_usage:模块辅助逻辑; write_and_return_billing_safety:写出文件或结果
scripts/run_pool_strategy_module.py	Run the first modular pool strategy against validate/stage/paper endpoints.	类 ScenarioQuoteSource(__init__, get_quotes); parse_args:解析命令行参数; main:命令入口; run_module:运行流程; build_source:构建对象/报告; build_universe_selection:构建对象/报告; monitor_symbols_from_records:监控状态; scenario_price_series:模块辅助逻辑; default_price_series:模块辅助逻辑; endpoint_for_mode:模块辅助逻辑; submit_order:提交请求; order_payload:模块辅助逻辑; decision_payload:模块辅助逻辑; compact_scan_report:模块辅助逻辑; pnl_snapshot:模块辅助逻辑; realized_pnl_for_decision:模块辅助逻辑; require_gateway_mode:模块辅助逻辑; request_json:发起请求; parse_symbols:模块辅助逻辑; parse_list:模块辅助逻辑; read_required:读取输入; read_trade_session_token:读取输入; payload_without_token:模块辅助逻辑; quote_payload:模块辅助逻辑; write_report:写出文件或结果; write_dashboard:写出文件或结果; elapsed_ms:模块辅助逻辑
scripts/run_position_protection.py	项目文件	main:命令入口
scripts/run_realtime_account_sync.py	Run realtime account state bus/snapshot/BUY-SELL sync reports.	main:命令入口
scripts/run_strategy_simulation.py	Run a local paper-only strategy simulation.	parse_args:解析命令行参数; main:命令入口
scripts/send_daily_feishu_update.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/send_daily_feishu_update_if_due.command	macOS 可双击/终端运行脚本	log:模块辅助逻辑
scripts/send_feishu_report.py	Send a Markdown/text report to a Feishu custom bot webhook.	parse_args:解析命令行参数; load_env_file:读取/加载数据; sanitize_text:模块辅助逻辑; clean_display_text:模块辅助逻辑; report_to_post_content:模块辅助逻辑; build_text_payload:构建对象/报告; build_post_payload:构建对象/报告; sign_payload:模块辅助逻辑; main:命令入口
scripts/serve_live_dashboard.py	Serve the IBKR read-only dashboard from localhost.	parse_args:解析命令行参数; parse_symbols:模块辅助逻辑; 类 LiveDataCache(__init__, get, _with_cache_status); fetch_data:拉取数据; next_client_id:模块辅助逻辑; build_system_status:构建对象/报告; fetch_trading_api_health:拉取数据; token_present:模块辅助逻辑; token_present_file_from_env:模块辅助逻辑; token_present_file:模块辅助逻辑; read_text_file:读取输入; make_handler:模块辅助逻辑; main:命令入口
scripts/start_dev_lock_api.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/start_live_dashboard.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/start_trade_lock_api.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/stop_trading_api.command	macOS 可双击/终端运行脚本	脚本式流程，无显式 shell function
scripts/submit_single_paper_from_readiness.py	Gate a single explicit paper limit order from readiness artifacts.	无顶层类/函数，主要为常量、导入或包初始化
scripts/test_ibkr_tick_by_tick.py	Sample IBKR tick-by-tick read-only market data.	类 TickByTickClient(__init__, nextValidId, error, tickByTickAllLast, tickByTickBidAsk, tickByTickMidPoint, _append, _event_time); parse_args:解析命令行参数; main:命令入口; stock_contract:模块辅助逻辑
scripts/test_streaming_quotes.py	Read-only smoke test for IBKR Level I streaming quotes.	parse_args:解析命令行参数; main:命令入口

strategies

文件	作用	关键函数/类
strategies/__init__.py	项目文件	无顶层类/函数，主要为常量、导入或包初始化
strategies/base.py	项目文件	类 MarketData(last, midpoint); 类 PortfolioState(position_for); 类 Signal
strategies/dual_ma.py	项目文件	类 DualMovingAverageStrategy(__init__, generate_signal)
strategies/grid.py	项目文件	类 GridStrategy(__init__, generate_signal)
strategies/mean_reversion.py	项目文件	类 ZScoreMeanReversionStrategy(__init__, generate_signal)
strategies/ml_baseline.py	项目文件	类 LightGbmStyleBaselineStrategy(__init__, generate_signal)

tests

文件	作用	关键函数/类
tests/__init__.py	Tests for the trading safeguards.	无顶层类/函数，主要为常量、导入或包初始化
tests/test_account_reports.py	项目文件	类 AccountReportsTests(test_daily_summary_counts_open_orders)
tests/test_account_state_manager.py	项目文件	类 AccountStateManagerTests(test_state_schema_dataclasses_have_required_fields, test_stale_state_detection, test_account_state_report_generation_from_cached_inputs, test_streaming_quote_cache_feeds_account_state_manager, test_quote_state_splits_bid_ask_last_and_use_case_threshold, test_sell_execution_requires_fresh_bid_and_buy_requires_fresh_ask, test_reports_may_use_last_but_execution_blocks_without_bid, test_streaming_bid_ask_gap_uses_market_session_classifier, test_execution_requires_force_refresh_and_blocks_failure, test_require_fresh_state_for_execution_blocks_stale_state, +2)
tests/test_api_service.py	项目文件	类 FakeSettings; 类 FakeService(__init__, submit_limit); 类 FailingService(submit_limit); 类 CapturingHandler(__init__, send_response, send_header, end_headers); 类 ApiServiceTests(test_stage_limit_route_submits_without_transmission, test_paper_limit_route_submits_with_transmission, test_error_response_includes_workflow_step)
tests/test_auto_open_pipeline.py	项目文件	类 AutoOpenPipelineTests(test_defaults_enable_discovery_news_pool_intent_and_simulation_safely, test_unavailable_ibkr_news_does_not_fail_pipeline, test_scanner_news_and_fast_guidance_cannot_submit_orders, test_candidates_pass_pool_layers_and_have_scores, test_simulation_creates_only_local_simulated_orders, test_paper_execution_interface_defaults_false, test_safety_config_rejects_real_execution_flags, test_buy_sell_share_same_cycle_snapshot_refs)
tests/test_autonomous_agent.py	项目文件	类 AutonomousAgentTests(lifecycle_for, test_mode9_market_closed_waits_without_execution, test_mode9_no_quotes_waits_without_execution, test_mode9_stale_ask_waits_without_execution, test_mode9_fresh_quotes_become_paper_ready, test_mode9_disconnected_waits_without_execution, test_mode9_ready_quotes_but_order_gate_blocked_by_risk, test_mode9_api_failure_is_recoverable_api_not_ready, test_quote_readiness_requires_fresh_bid_and_ask, test_waiting_lifecycle_states_do_not_run_strategy, +7)
tests/test_autonomous_runtime.py	项目文件	类 FakeScanner(__init__, scan_once); 类 AutonomousRuntimeTests(test_kill_switch_blocks_runtime, test_tws_health_failure_pauses_runtime, test_runtime_submits_session_approved_bracket_order, test_runtime_blocks_event_without_protective_stop, test_live_order_remains_disabled); _runtime:内部辅助; _ready_health:内部辅助; _event:内部辅助
tests/test_candidate_research.py	项目文件	类 CandidateResearchTests(test_hard_audit_approves_structured_candidate, test_hard_audit_rejects_low_confidence, test_hard_audit_rejects_existing_universe_symbol, test_approval_upserts_candidate_row); candidate:模块辅助逻辑
tests/test_daily_feishu_update.py	项目文件	类 DailyFeishuUpdateTests(test_status_lines_include_repair_mode_and_blocked_reason, test_zero_submission_explanation_names_repair_state)
tests/test_dashboard.py	项目文件	类 DashboardTests(test_dashboard_contains_core_sections, test_dashboard_can_auto_refresh_and_show_missing_symbols)
tests/test_event_risk_control.py	项目文件	类 EventRiskControlTests(test_high_event_blocks_buy_and_requires_outside_rth_protection)
tests/test_event_router.py	项目文件	类 EventRouterTests(test_routes_report_only_actions_from_risk_reports)
tests/test_full_paper_trial_pipeline.py	项目文件	类 PullbackQuoteSource(__init__, get_quotes); 类 FullPaperTrialPipelineTests(test_full_pipeline_writes_all_reports_without_submitting_paper)
tests/test_gap_escape_manager.py	项目文件	类 GapEscapeManagerTests(test_trigger_when_bid_below_stop_limit_and_marketable_limit_price, test_blocked_when_quote_stale, test_blocked_when_spread_too_wide); _settings:内部辅助
tests/test_gap_risk_manager.py	项目文件	类 GapRiskManagerTests(test_worst_gap_loss_and_max_notional, test_over_budget_recommends_reduce_or_hedge, test_event_risk_increases_gap_shock)
tests/test_guarded_paper_trial_session.py	项目文件	类 PullbackQuoteSource(__init__, get_quotes); guarded_config:模块辅助逻辑; 类 GuardedPaperTrialSessionTests(test_blocks_before_pipeline_when_environment_is_not_ready, test_ready_environment_runs_full_pipeline_without_paper_submit_by_default)

文件	作用	关键函数/类
tests/test_ibkr_callback_bridge.py	项目文件	类 IbkrCallbackBridgeTests(test_data_bus_architecture_has_four_logical_buses, test_bootstrap_loads_cache_then_waits_for_real_ibkr, test_real_ibkr_callback_writes_to_bus_and_replaces_cache, test_older_real_ibkr_event_is_stored_but_does_not_overwrite_newer, test_stale_field_remains_subscribed_and_new_event_returns_fresh, test_ask_freshness_independent_from_other_domains, test_dry_run_uses_no_orders_or_snapshots, test_wiring_audit_reports_bridge_separately_from_real_events)
tests/test_ibkr_callback_wiring_audit.py	项目文件	类 IbkrCallbackWiringAuditTests(setUp, test_report_distinguishes_callback_registration_from_freshness, test_report_distinguishes_real_ibkr_simulation_fixture_and_none_sources, test_stale_ask_does_not_mark_bus_failed, test_stale_ask_blocks_real_paper_buy_but_not_local_simulation, test_safety_flags_remain_zero_or_false)
tests/test_ibkr_mean_reversion_pipeline.py	项目文件	类 OscillatingQuoteSource(__init__, get_quotes); 类 IbkrMeanReversionPipelineTests(test_pipeline_records_quotes_and_writes_diagnostics)
tests/test_ibkr_quote_recorder.py	项目文件	类 FakeQuoteSource(__init__, get_quotes); 类 IbkrQuoteRecorderTests(test_record_quotes_writes_research_csv_and_report, test_record_quotes_requires_symbols)
tests/test_ibkr_readonly.py	项目文件	类 IbkrReadOnlyQuoteClientTests(test_tick_callbacks_build_normalized_quote, test_close_price_can_create_quote_when_last_is_missing, test_delayed_tick_callbacks_build_normalized_quote, test_simulation_source_can_build_ibkr_readonly_provider, test_stock_contract_can_target_iex, test_parallel_source_chunks_symbols_without_tws_connection)
tests/test_ibkr_streaming.py	项目文件	类 FakeStreamingClient(__init__, connect, run_loop, reqMarketDataType, reqMktData, cancelMktData, isConnected, disconnect); 类 IbkrStreamingQuoteTests(test_req_mkt_data_uses_streaming_snapshot_disabled, test_more_than_max_symbols_is_truncated_safely, test_tick_price_updates_quote_cache, test_bid_ask_midpoint_can_create_usable_quote, test_stale_quote_is_detected, test_shutdown_cancels_subscriptions, test_streaming_errors_are_recorded_per_symbol, test_streaming_module_does_not_call_order_api)
tests/test_market_data_billing_safety.py	项目文件	类 MarketDataBillingSafetyTests(test_snapshot_true_is_blocked_for_automatic_quote_acquisition, test_missing_quote_does_not_trigger_paid_or_regulatory_snapshot, test_delayed_and_frozen_data_cannot_pass_execution_readiness, test_live_fresh_bid_and_ask_can_pass_quote_readiness, test_reports_can_use_last_fallback_but_execution_cannot, test_market_data_billing_safety_report_schema, test_force_refresh_dry_run_submits_and_cancels_no_orders)
tests/test_market_quote_crosscheck.py	项目文件	类 MarketQuoteCrosscheckTests(test_closed_market_missing_bid_ask_is_normal, test_active_market_missing_bid_ask_is_quote_feed_problem, test_subscription_reason_is_subscription_problem, test_paper_automation_sqlite_tables_exist)
tests/test_market_session.py	项目文件	类 MarketSessionTests(test_regular_session_expects_live_bid_ask, test_premarket_session_expects_live_bid_ask, test_afterhours_session_expects_live_bid_ask, test_after_afterhours_close_blocks_quote_wait, test_weekend_blocks_quote_wait, test_holiday_blocks_quote_wait, test_early_close_active_then_closed_after_1300, test_symbol_rows_report_contract_details_when_provided, test_contract_details_closed_overrides_calendar_open_for_symbol, test_unknown_calendar_blocks_execution_when_verified_calendar_required, +2)
tests/test_mean_reversion_diagnostics.py	项目文件	类 MeanReversionDiagnosticsTests(test_oscillating_series_is_research_candidate, test_trending_series_is_rejected, test_csv_reader_groups_prices_by_symbol, test_write_report_creates_json_artifact)
tests/test_mode9_infrastructure_status.py	项目文件	类 Mode9InfrastructureStatusTests(test_feature_status_schema_contains_required_fields, test_sqlite_tables_exist_after_ensure)
tests/test_mode9_report_only_infrastructure_cycle.py	项目文件	类 Mode9ReportOnlyInfrastructureCycleTests(test_cycle_is_report_only_and_calls_core_modules)
tests/test_monitoring_line_control.py	项目文件	类 MonitoringLineControlTests(test_monitor_on_keeps_safety_flags_conservative, test_market_session_and_rollout_precheck_commands_exist, test_full_paper_automation_command_keeps_hard_safety_flags, test_mode9_runtime_ensures_trade_lock_before_agent, test_interactive_menu_folds_advanced_controls, test_mode9_runtime_keeps_hard_safety_flags, test_mode9_runtime_reuses_existing_token_or_restarts_api)
tests/test_openclaw_client.py	项目文件	类 OpenClawTradingClientTests(test_paper_submitted_is_success, test_pending_confirmation_is_not_reported_as_success, test_rejected_order_is_not_reported_as_success, test_validation_rejection_uses_rejected_exit_code, test_unknown_order_status_is_not_reported_as_success, test_client_timing_is_added_to_result, test_preflight_rejects_non_trade_lock, test_preflight_rejects_tws_not_ready, test_api_connection_failure_reports_client_http, test_fast_paper_limit_skips_client_preflight)
tests/test_options_hedge_planner.py	项目文件	类 OptionsHedgePlannerTests(test_not_triggered_below_notional_threshold, test_triggered_above_notional_threshold, test_quantity_under_100_blocks_single_stock_option_hedge); _settings:内部辅助
tests/test_order_intents_and_pnl_ledger.py	项目文件	类 OrderIntentAndPnlLedgerTests(test_buy_intent_requires_buy_side, test_options_plan_cannot_be_submitted_in_intent_task, test_structured_pnl_tables_have_queryable_columns, test_intents_route_to_split_ledgers_without_submissions)
tests/test_paper_candidate_hunt.py	项目文件	类 SequenceQuoteSource(__init__, get_quotes); 类 PaperCandidateHuntTests(test_hunt_stops_when_validate_payload_is_found, test_hunt_reports_static_quotes_without_forcing_order)

文件	作用	关键函数/类
tests/test_paper_environment_audit.py	项目文件	make_settings:模块辅助逻辑; 类 PaperEnvironmentAuditTests(test_ready_report_requires_trade_lock_and_tws_paper_account, test_dry_run_without_health_is_blocked_with_next_steps, test_unready_tws_blocks_even_when_local_settings_are_paper)
tests/test_paper_execution_gate.py	项目文件	类 PaperExecutionGateTests(test_build_gate_report_selects_approved_one_share_candidate, test_gate_blocks_without_ready_readiness, test_execute_requires_confirmation_phrase, test_execute_posts_paper_limit_after_health_passes, test_execute_accepts_matching_pre_submit_review, test_execute_blocks_stale_pre_submit_review, test_execute_blocks_pre_submit_review_payload_mismatch)
tests/test_paper_pre_submit_review.py	项目文件	类 PaperPreSubmitReviewTests(test_ready_review_requires_gate_order_sn_apshot_and_pnl, test_open_order_blocks_review)
tests/test_paper_readiness_report.py	项目文件	类 PaperReadinessReportTests(test_report_requires_validate_before_paper, test_report_marks_manual_review_after_all_validates_pass, test_load_and_write_report_round_trip)
tests/test_paper_session_runbook.py	项目文件	类 PaperSessionRunbookTests(test_blocked_runbook_lists_blockers_and_safe_commands, test_ready_runbook_includes_validate_and_explicit_paper_commands, test_markdown_and_json_are_written)
tests/test_paper_session_status_page.py	项目文件	类 PaperSessionStatusPageTests(test_render_status_page_contains_core_sections, test_write_status_page_creates_html_file)
tests/test_paper_trial_evidence_bundle.py	项目文件	类 PaperTrialEvidenceBundleTests(test_blocked_environment_classifies_before_paper_window, test_submitted_without_reconciliation_needs_reconciliation, test_ready_execution_gate_classifies_explicit_paper_submit, test_reconciled_stage_requires_reconciled_report, test_reviewed_gate_reconciled_stage_does_not_require_guarded_session_submission, test_writers_create_json_and_markdown)
tests/test_paper_trial_preflight_checklist.py	项目文件	类 PaperTrialPreflightChecklistTests(test_ready_checklist_allows_validate_only_window, test_blocked_audit_blocks_real_ibkr_window, test_missing_rehearsal_requires_attention_after_ready_audit)
tests/test_paper_trial_readiness_monitor.py	项目文件	类 PaperTrialReadinessMonitorTests(test_monitor_waits_without_quote_or_order_side_effects, test_monitor_stops_when_preflight_becomes_validate_ready)
tests/test_paper_trial_reconciliation.py	项目文件	类 PaperTrialReconciliationTests(test_missing_gate_is_no_paper_submission, test_unsubmitted_gate_is_no_paper_submission, test_submitted_gate_reconciles_required_readbacks, test_submitted_gate_without_tws_readback_is_incomplete, test_write_reconciliation_creates_json)
tests/test_paper_trial_rehearsal.py	项目文件	类 PaperTrialRehearsalTests(test_rehearsal_runs_ready_path_without_ibkr_or_orders)
tests/test_paper_trial_report_refresh.py	项目文件	类 FailingQuoteSource(get_quotes); 类 PaperTrialReportRefreshTests(test_refresh_blocked_environment_writes_reports_without_quote_source, test_refresh_reads_audit_record_for_submitted_gate_reconciliation)
tests/test_paper_validation_plan.py	项目文件	类 PaperValidationPlanTests(test_negative_z_candidate_creates_buy_validate_payload, test_positive_z_candidate_is_watch_only_for_long_module, test_rejected_diagnostics_do_not_create_payload, test_load_and_write_plan_round_trip)
tests/test_pool_architecture_audit.py	项目文件	类 PoolArchitectureAuditTests(test_audit_reports_six_layer_runtime_wired_but_execution_gated)
tests/test_pool_manager.py	项目文件	类 PoolManagerTests(test_builds_six_layers_and_forces_positions_into_monitor_pool)
tests/test_pool_state.py	项目文件	类 PoolStateTests(manager, add_ready_watch, test_normal_I1_to_I6_promotion, test_ineligible_symbol_remains_outside_I2, test_stale_ask_blocks_I5_to_I6, test_market_closed_blocks_unsupported_execution, test_spread_widening_demotes_watch_pool_symbol, test_expired_signal_is_removed, test_expired_news_event_cannot_authorize_trade, test_duplicate_order_is_blocked, +7)
tests/test_position_protection.py	项目文件	类 PositionProtectionTests(fresh_ok, test_missing_stop_detection, test_underprotected_detection, test_fully_protected_detection, test_overprotected_detection, test_duplicate_stop_detection, test_protective_sell_stop_quantity_calculation, test_stop_price_normalization, test_current_price_hard_stop_when_market_below_avg_stop, test_missing_stop_generates_repair_proposal, +30); _settings:内部辅助; _guard_row:内部辅助; _stop_config:内部辅助
tests/test_position_protection_test_result_refresh.py	项目文件	类 PositionProtectionTestResultRefreshTests(test_plain_stp_outside_rth_false_is_success_with_warning)
tests/test_price_normalizer.py	项目文件	类 PriceNormalizerTests(test_buy_limit_floors_to_cent, test_sell_limit_ceils_to_cent, test_sell_stop_floors_to_valid_protective_stop, test_buy_stop_ceils_to_cent, test_custom_tick_size)
tests/test_profit_protection_reports.py	项目文件	类 ProfitProtectionReportTests(test_profit_lock_take_profit_and_trailing_are_report_only)
tests/test_project_completion_audit.py	项目文件	类 ProjectCompletionAuditTests(test_current_blocked_state_is_not_complete, test_reconciled_stage_can_complete_audit, test_writers_create_audit_artifacts, test_validate_attempt_without_payload_is_not_complete)
tests/test_quote_quality_report.py	项目文件	类 QuoteQualityReportTests(test_static_last_prices_are_flagged, test_moving_prices_are_usable, test_writers_create_artifacts, _write_rows)

文件	作用	关键函数/类
tests/test_realtime_account_sync.py	项目文件	类 RealtimeAccountSyncTests(test_realtime_bus_accepts_state_events, test_cycle_snapshot_captures_latest_state_versions, test_buy_and_sell_decisions_share_same_snapshot_id, test_high_priority_sell_blocks_same_symbol_buy, test_freshness_blocks_state_quote_pnl_and_position, test_audit_writer_is_single_writer_and_wal_enabled_append_only, test_buy_event_has_pending_protection_fields, test_position_lifecycle_state_is_written, test_strategies_and_engines_cannot_submit_orders, test_end_to_end_reports_no_real_order_or_snapshot_risk)
tests/test_risk.py	项目文件	类 RiskEngineTests(setUp, test_disarmed_engine_rejects_every_order, test_valid_paper_proposal_is_approved_when_armed, test_order_over_value_limit_is_rejected, test_order_over_quantity_limit_is_rejected)
tests/test_security_master.py	项目文件	类 SecurityMasterTests(test_builds_security_master_from_universe_csv, test_persistent_security_master_takes_precedence_over_seed_universe, test_nasdaq_trader_rows_build_broad_master_records)
tests/test_service.py	项目文件	make_settings:模块辅助逻辑; valid_payload:模块辅助逻辑; valid_limit_payload:模块辅助逻辑; ready_tws_status:读取输入; not_ready_tws_status:模块辅助逻辑; 类 TradingServiceTests(test_stage_me nu_requires_explicit_confirmation, test_dry_run_validation_approves_safe_proposal, test_dry_run_mode_cannot_stage_order, test_paper_transmission_requires_explicit_switch, test_kill_switch_rejects_submission, test_paper_transmission_requires_session_token_configured, test_paper_transmission_rejects_invalid_session_token, test_paper_transmission_accepts_valid_session_token, test_duplicate_idempotency_key_is_rejected, test_outside_rth_switch_is_passed_to_broker, +10)
tests/test_session_calendar.py	项目文件	类 SessionCalendarTests(test_regular_session_classification, test_premarket_session_classification, test_afterhours_session_classification, test_overnight_session_classification, test_weekend_offline_classification)
tests/test_session_policy.py	项目文件	类 SessionPolicyTests(test_outside_rth_only_for_extended_sessions, test_protective_stop_required, test_stale_quote_blocks_order, test_open_order_blocks_duplicate_submit, test_max_orders_per_symbol_blocks_order); _quote:内部辅助
tests/test_strategy.py	项目文件	类 StrategyFrameworkTests(test_value_filter_approves_quality_candidate, test_value_filter_rejects_weak_candidate, test_value_filter_scores_undervaluation_quality_demand_and_safety, test_moving_average_requires_confirmation_before_entry, test_tactical_strategy_blocks_stale_quote, test_tactical_strategy_creates_bracket_style_plan, test_quote_cache_tracks_freshness, test_ring_buffer_keeps_only_recent_values, test_rotating_symbol_pool_cycles_fixed_size_batches, test_rotating_strategy_scanner_updates_cache_and_bar_ring, +1)
tests/test_strategy_catalog.py	项目文件	类 StrategyCatalogTests(test_catalog_covers_current_strategy_modules_and_books, test_writers_create_catalog_artifacts)
tests/test_strategy_extension_status.py	项目文件	类 StrategyExtensionStatusTests(test_status_is_report_only_and_does_not_enable_live_or_buy, test_write_reports_creates_json_and_markdown); _settings:内部辅助
tests/test_strategy_modules.py	项目文件	类 ConservativeTrendModuleTests(test_module_buys_confirmed_classic_uptrend, test_module_sells_after_profit_target, test_module_blocks_unapproved_symbol, test_portfolio_risk_gate_blocks_extra_entry_after_position_limit, test_portfolio_risk_gate_blocks_daily_loss); bars_for:模块辅助逻辑
tests/test_strategy_validation_plan.py	项目文件	类 StrategyValidationPlanTests(test_grid_pullback_creates_validate_payload, test_wide_spread_rejects_buy_signal_before_validate); _write_quotes:内部辅助
tests/test_tws_paper.py	项目文件	类 TwsPaperBrokerTests(test_us_stock_price_normalizes_to_cent_tick, test_transmitted_order_requires_order_status_ack, test_staged_order_can_ack_from_open_order, test_bracket_has_protective_stop_and_no_transmission_by_default, test_transmitted_bracket_only_transmits_on_final_stop_child, test_bracket_child_failure_requests_parent_cancel, test_outside_rth_is_configurable_for_both_bracket_legs, test_limit_order_can_transmit_outside_rth_without_stop_leg, test_stop_limit_order_is_sell_gtc_outside_rth_with_order_ref, test_order_confirmation_records_partial_ack_as_pending, +1)
tests/test_unified_strategy_interface.py	项目文件	类 UnifiedStrategyInterfaceTests(test_dual_ma_generates_buy_signal, test_zscore_mean_reversion_generates_buy_signal, test_grid_generates_sell_signal_for_upper_band, test_ml_baseline_uses_precomputed_probability, test_risk_and_paper_order_path, test_live_order_is_disabled_by_default)
tests/test_universe.py	项目文件	类 UniverseTests(test_default_universe_loads_large_enabled_pool, test_select_universe_intersects_gateway_allowlist, test_symbols_csv_deduplicates_and_normalizes, test_loader_respects_disabled_rows)

trading

文件	作用	关键函数/类
trading/_/_init__.py	Core types and safeguards for the trading service.	无顶层类/函数，主要为常量、导入或包初始化

文件	作用	关键函数/类
trading/account_state_manager.py	项目文件	build_account_state:构建对象/报告; force_refresh_state:模块辅助逻辑; build_force_refresh_dry_run:构建对象/报告; refresh_streaming_quote_diagnostics:刷新状态; overlay_streaming_quotes:模块辅助逻辑; symbols_from_guard:模块辅助逻辑; require_fresh_state_for_execution:模块辅助逻辑; build_quote_states:构建对象/报告; quote_state_from_guard:模块辅助逻辑; quote_execution_readiness:模块辅助逻辑; quote_stale_reason:模块辅助逻辑; actual_quote_age_for_use_case:模块辅助逻辑; quote_age_sec:模块辅助逻辑; quote_threshold:模块辅助逻辑; parse_market_data_type:模块辅助逻辑; market_data_type_label:模块辅助逻辑; quote_blocked_reason:模块辅助逻辑; position_state_from_guard:模块辅助逻辑; open_order_state:模块辅助逻辑; coverage_state:模块辅助逻辑; build_pnl_states:构建对象/报告; pnl_subscription_support:模块辅助逻辑; build_freshness:构建对象/报告; stale_status:模块辅助逻辑; write_account_state_reports:写出文件或结果; account_state_payload:模块辅助逻辑; account_dashboard_payload:模块辅助逻辑; account_execution_readin
trading/audit.py	项目文件	类 AuditLog(__init__, _connect, _initialize, reserve, update, get, list_recent, list_for_date, paper_notional_for_date, _row_to_record); _notional:内部辅助
trading/audit_writer.py	Single-writer audit queue for SQLite append-only ledgers.	类 AuditEvent; 类 AuditWriter(__init__, submit, submit_event_store, flush, queued_count)
trading/auto_open_pipeline.py	Auto-open local discovery, news, dynamic pool, and simulation pipeline.	类 PipelineConfig(from_env); run_auto_open_pipeline:运行流程; assert_simulation_safety:模块辅助逻辑; run_discovery_orchestrator:运行流程; run_ibkr_scanner:运行流程; run_news_pipeline:运行流程; news_provider_status:模块辅助逻辑; ibkr_news_interface_tests:模块辅助逻辑; raw_news_fixture:模块辅助逻辑; structure_news_event:模块辅助逻辑; run_dynamic_pool:运行流程; run_fast_order_guidance:运行流程; run_order_intent_router:运行流程; run_local_simulation:运行流程; run_paper_execution_gate:运行流程; write_final_audit:写出文件或结果; consolidate_candidates:模块辅助逻辑; score_candidate:计算评分; add_pool_membership:模块辅助逻辑; write_report:写出文件或结果; load_universe_rows:读取/加载数据; current_position_symbols:模块辅助逻辑; open_order_symbols_from_reports:模块辅助逻辑; research_symbols:模块辅助逻辑; cached_scanner_symbols:模块辅助逻辑; scanner_symbols_for_code:模块辅助逻辑; env_symbols:模块辅助逻辑; tws_available:模块辅助逻辑; market_session_state:模块辅助逻辑; simulated_price:模块辅助逻辑
trading/autonomous_runtime.py	项目文件	类 Scanner(scan_once); 类 AutonomousRuntimeConfig(from_env); 类 RuntimeCycleReport; 类 AutonomousPaperRuntime(__init__, run_once, run_forever, _cycle, _write_status, _append_events); build_runtime:构建对象/报告; _read_open_orders:内部辅助; _report:内部辅助; _health_ready:内部辅助; _proposal_from_event:内部辅助; quote_from_event:内部辅助; _float_or_none:内部辅助; _bool_env:内部辅助
trading/buy_decision_engine.py	BUY decision engine. Emits BUY intents only; never calls IBKR directly.	build_buy_intent:构建对象/报告; can_submit_order:模块辅助逻辑; submit_order:提交请求
trading/candidate_research.py	项目文件	类 Evidenceltem; 类 LimPreReview; 类 CandidateDraft; 类 HardAuditConfig; 类 HardAuditResult; 类 CandidateReport(as_dict); build_candidate_report:构建对象/报告; audit_candidate:审计状态; normalize_draft:标准化数据; load_candidate_draft:读取/加载数据; candidate_draft_from_dict:模块辅助逻辑; write_candidate_report:写出文件或结果; candidate_report_from_dict:模块辅助逻辑; load_candidate_report:读取/加载数据; _profile_from_dict:内部辅助; _pre_review_from_dict:内部辅助; _evidence_from_dict:内部辅助; _tuple_of_strings:内部辅助; _float_or_none:内部辅助; _int_or_none:内部辅助; _bool_or_true:内部辅助; _bool_or_none:内部辅助
trading/candidate_scoring.py	Candidate scoring helpers.	score:计算评分
trading/config.py	项目文件	_read_bool:内部辅助; _load_or_create_api_key:内部辅助; _load_optional_file_value:内部辅助; _load_allowed_symbols:内部辅助; _symbols_from_file:内部辅助; _read_symbol_tuple:内部辅助; 类 Settings(load); _read_optional_float:内部辅助
trading/cycle_snapshot.py	Immutable decision-time snapshots over the realtime account state bus.	类 CycleSnapshot; create_cycle_snapshot:创建对象; write_report:写出文件或结果
trading/dashboard.py	项目文件	render_dashboard_html:渲染输出
trading/data_buses.py	Logical data bus architecture for realtime decisions.	类 MarketDataBus(__init__, update_quote); 类 NewsTriggerBus(__init__, update); 类 DiscoveryPoolBus(__init__, update); build_data_bus_architecture_report:构建对象/报告
trading/discovery_orchestrator.py	Unified discovery orchestrator endpoint.	run:运行流程
trading/event_risk_control.py	项目文件	类 EventRiskConfig; load_event_risk_config:读取/加载数据; build_event_risk_report:构建对象/报告; recommended_event_action:模块辅助逻辑; load_local_events:读取/加载数据; write_report:写出文件或结果; _event_for_symbol:内部辅助; _latest_net_liquidation:内部辅助; _bool_env:内部辅助; _float_env:内部辅助; _now:内部辅助
trading/event_router.py	项目文件	build_event_router_report:构建对象/报告; event:模块辅助逻辑; write_report:写出文件或结果; read_json:读取输入
trading/execution_ledger_split.py	Build split BUY/SELL intent ledgers and structured PnL reports.	build_execution_and_pnl_ledgers:构建对象/报告; buy_intents:模块辅助逻辑; protective_sell_intents:模块辅助逻辑; gap_escape_intents:模块辅助逻辑; profit_sell_intents:模块辅助逻辑; event_risk_sell_intents:模块辅助逻辑; _sell_plan_intents:内部辅助; options_plan_intents:模块辅助逻辑; persist_intents:模块辅助逻辑; build_pnl_ledger_report:构建对象/报告; persist_pnl:模块辅助逻辑; insert_direct:模块辅助逻辑; attribution:模块辅助逻辑; execution_split_report:模块辅助逻辑; separation_report_payload:模块辅助逻辑; count_intents:模块辅助逻辑; write_reports:写出文件或结果; markdown:模块辅助逻辑; read_json:读取输入; held_symbols:模块辅助逻辑; read_market_session_state:读取输入; count_values:模块辅助逻辑; to_float:转换输出
trading/fast_order_guidance.py	Fast order guidance from scanner/news events. Guidance only; no order submission.	run:运行流程

文件	作用	关键函数/类
trading/gap_escape_manager.py	项目文件	类 GapEscapeConfig; load_gap_escape_config:读取/加载数据; build_gap_escape_report:构建对象/报告; gap_escape_row:模块辅助逻辑; gap_escape_blocked_reason:模块辅助逻辑; _trigger_reason:内部辅助; _active_stop_price:内部辅助; _active_stop_limit_price:内部辅助; _worst_loss_by_symbol:内部辅助; write_report:写出文件或结果; _bool_env:内部辅助; _float_env:内部辅助; _float:内部辅助; _now:内部辅助
trading/gap_risk_manager.py	项目文件	类 GapRiskConfig; load_gap_risk_config:读取/加载数据; build_gap_risk_report:构建对象/报告; gap_risk_row:模块辅助逻辑; classify_symbol_shock:分类状态; write_report:写出文件或结果; _long_positions:内部辅助; _latest_net_liquidation:内部辅助; _dedupe_shocks:内部辅助; _float:内部辅助; _float_env:内部辅助; _now:内部辅助
trading/ibkr_callback_bridge.py	REAL_IBKR callback bridge into logical realtime data buses.	类 IbkrCallbackBridge(_init_, nextValidId, connectionClosed, error, tickPrice, tickSize, tickString, tickGeneric, marketDataType, tickReqParams, +18); run_ibkr_callback_dry_run:运行流程; stock_contract:模块辅助逻辑
trading/ibkr_callback_wiring_audit.py	Audit whether real IBKR callbacks are wired into RealtimeAccountStateBus.	build_ibkr_callback_wiring_audit:构建对象/报告; callback_row:模块辅助逻辑; bus_state_sources:模块辅助逻辑; callback_defined:模块辅助逻辑; callback_wires_bus:模块辅助逻辑; latest_bus_timestamp_for_callback:模块辅助逻辑; callback_notes:模块辅助逻辑; callback_wired:模块辅助逻辑; tws_socket_available:模块辅助逻辑; update_realtime_reports:更新状态; write_report:写出文件或结果; read_json:读取输入; utc_now:模块辅助逻辑
trading/ibkr_news.py	IBKR news interface diagnostics.	run_interface_tests:运行流程
trading/ibkr_readonly.py	项目文件	类 _QuoteState(to_quote); 类 IbkrReadOnlyQuoteClient(_init_, nextValidId, run_loop, error, tickPrice, tickSize, tickSnapshotEnd, _state, _mark_ready_if_possible); 类 IbkrReadOnlyQuoteSource(_init_, get_quotes, _stock_contract, _wait_for_quotes); 类 ParallelIbkrReadOnlyQuoteSource(_init_, get_quotes, _source_for); _chunks:内部辅助; _env_bool:内部辅助
trading/ibkr_scanner.py	IBKR scanner diagnostics and local discovery fallback.	run:运行流程
trading/ibkr_streaming.py	项目文件	类 StreamingQuoteState(to_quote, age_ms, field_age_ms, market_data_type_name); 类 IbkrStreamingQuoteClient(_init_, nextValidId, run_loop, error, tickPrice, tickSize, tickString, marketDataType, _state); 类 IbkrStreamingQuoteSource(_init_, start, stop, get_latest_quotes, get_quote, quote_age_ms, is_stale, stale_symbols, errors, active, +3); _clean_symbols:内部辅助; _log_event:内部辅助; streaming_blocked_reason:模块辅助逻辑
trading/market_data.py	项目文件	类 Quote(age_ms, midpoint, spread); 类 Bar; 类 QuoteSource(get_quotes); 类 InMemoryQuoteCache(_init_, update, update_many, get, is_fresh, snapshot); 类 RingBuffer(_init_, append, replace_latest, latest, values); 类 SymbolRingBuffers(_init_, append, replace_latest, latest, values, snapshot, _buffer); 类 RotatingSymbolPool(_init_, symbols, next_batch); 类 BarBuilder(_init_, update, active_bar, _new_bar); 类 SimulatedQuoteSource(_init_, get_quotes); 类 StooqDelayedQuoteSource(_init_, get_quotes, _row_to_quote); 类 YahooDelayedQuoteSource(_init_, get_quotes, _quote_from_chart_payload, _quotes_from_chart_payload); 类 YahooChartReplayQuoteSource(_init_, get_quotes, _load_symbol); quote_stream:模块辅助逻辑; _bucket_start:内部辅助; _float_or_none:内部辅助; _int_or_none:内部辅助
trading/market_data_line_manager.py	项目文件	类 MarketDataLineConfig; 类 MarketDataLinePlan; 类 MarketDataLineManager(_init_, plan, write_report); _unique:内部辅助
trading/market_quote_crosscheck.py	Cross-validate market session expectations against received quotes.	build_market_quote_crosscheck_report:构建对象/报告; crosscheck_row:模块辅助逻辑; classify_cross_validation_result:分类状态; summarize:汇总信息; write_report:写出文件或结果; _markdown:内部辅助; _read_json:内部辅助; _now:内部辅助
trading/market_session.py	US equity market-session classification and reporting.	类 MarketSessionState; 类 MarketSessionConfig; 类 CalendarStatus; 类 MarketSessionSnapshot(to_dict); config_from_env:模块辅助逻辑; current_market_session:模块辅助逻辑; write_market_session_report:写出文件或结果; fetch_ibkr_contract_details:拉取数据; market_session_md:模块辅助逻辑; classify_streaming_bid_ask_gap:分类状态; parse_ibkr_trading_hours:模块辅助逻辑; 类 _ContractDetailsClient(_init_, nextValidId, contractDetails, contractDetailsEnd, error); _stock_contract:内部辅助; calendar_status:模块辅助逻辑; _state_for_time:内部辅助; _symbol_rows:内部辅助; _hours_today:内部辅助; _hours_allow_now:内部辅助; _hhmm_to_minutes:内部辅助; _calendar_status_from_package:内部辅助; _calendar_status_from_local_file:内部辅助; _inactive_reason:内部辅助; _next_regular_open:内部辅助; _holiday_dates:内部辅助; _early_close_dates:内部辅助; _observed:内部辅助; _nth_weekday:内部辅助; _last_weekday:内部辅助; _good_friday:内部辅助; _time_env:内部辅助; _parse_time:内部辅助
trading/models.py	项目文件	类 TradeProposal; 类 RiskDecision
trading/module_interfaces.py	项目文件	类 StateFreshness; 类 QuoteState; 类 PnLState; 类 ProtectionCoverageState; 类 OpenOrderState; 类 PositionState; 类 ForceRefreshResult; 类 AccountState
trading/news_candidate_router.py	Route structured news into candidate/risk events.	run:运行流程
trading/news_event_monitor.py	News event monitor endpoint.	run:运行流程
trading/news_event_structurer.py	Normalize raw news events into structured news events.	normalize:标准化数据
trading/news_provider_registry.py	Unified news provider registry for IBKR, RSS, SEC, earnings, cache, and fixtures.	build_registry:构建对象/报告
trading/options_hedge_planner.py	项目文件	类 OptionHedgeConfig; load_option_hedge_config:读取/加载数据; build_options_hedge_report:构建对象/报告; option_hedge_plan:模块辅助逻辑; write_report:写出文件或结果; _markdown:内部辅助; _bool_env:内部辅助; _float_env:内部辅助; _now:内部辅助
trading/order_intent_router.py	Order intent router. Emits synchronized intent events only.	route_synchronized_intents:路由事件/候选; write_conflict_report:写出文件或结果; run:运行流程

文件	作用	关键函数/类
trading/order_intents.py	Unified order intent model for Mode 9 paper automation ledgers.	类 OrderIntent(to_dict, sqlite_values); make_intent_id:模块辅助逻辑; utc_now:模块辅助逻辑
trading/order_rejections.py	项目文件	log_order_rejection:模块辅助逻辑; rejection_result:模块辅助逻辑; _parse_ib_error:内部辅助
trading/paper_audit_db.py	SQLite logging helpers for Mode 9 paper automation.	ensure_paper_automation_tables:确保前置条件; insert_event:模块辅助逻辑; generic_decision_event:模块辅助逻辑; sqlite_table_summary:模块辅助逻辑; utc_now:模块辅助逻辑; _column_definitions:内部辅助
trading/paper_environment_audit.py	项目文件	类 PaperEnvironmentAuditItem; 类 PaperEnvironmentAuditReport; build_environment_audit:构建对象/报告; write_environment_audit:写出文件或结果; main:命令入口; fetch_health:拉取数据; _audit_items:内部辅助; _health_items:内部辅助; _item:内部辅助; _report_status:内部辅助; _next_steps:内部辅助; _settings_snapshot:内部辅助; _load_json_object:内部辅助; _dedupe:内部辅助
trading/paper_execution_gate.py	Future paper execution gate. Report-only and default false.	run:运行流程
trading/paper_order_executor.py	Future IBKR paper order executor interface. Disabled by default.	submit_order_if_enabled:提交请求
trading/pool_builder.py	Dynamic six-layer pool builder.	run:运行流程
trading/pool_manager.py	项目文件	类 PoolMembership; build_pool_manager_report:构建对象/报告; discovery_eligibility:发现候选; architecture_report:模块辅助逻辑; write_reports:写出文件或结果; read_json:读取输入
trading/pool_state.py	Canonical six-layer symbol pool state.	类 PoolTransition; 类 SymbolPoolState; 类 PoolStateManager(__init__, upsert_master, evaluate_eligibility, nominate_scan, promote_to_watch, evaluate_signal, enter_execution_pool, apply_order_callback, expire, save, +8); quote_readiness:模块辅助逻辑; signal_failures:模块辅助逻辑; expired_signal:模块辅助逻辑; has_active_order:布尔判断; normalize_status:标准化数据; state_to_dict:模块辅助逻辑; state_from_dict:模块辅助逻辑; normalize_symbol:标准化数据; to_float:转换输出; age_seconds:模块辅助逻辑; iso_add:模块辅助逻辑; parse_time:模块辅助逻辑
trading/position_guard.py	项目文件	类 PositionRecord; 类 ProtectionStatus; 类 ReadOnlyPositionsClient(__init__, nextValidId, managedAccounts, updatePortfolio, accountDownloadEnd, error, run_loop); read_positions:读取输入; build_position_guard_report:构建对象/报告; write_report:写出文件或结果; evaluate_protection_status:评估决策; _read_quotes:内部辅助; _read_streaming_quote_cache:内部辅助; _parse_timestamp:内部辅助; _float:内部辅助; _market_data_type_from_name:内部辅助; _env_bool:内部辅助; _recommended_action:内部辅助; _order_details:内部辅助
trading/position_lifecycle_manager.py	Position lifecycle manager for simulation, paper, and live-forbidden modes.	类 PositionLifecycle; 类 PositionLifecycleManager(__init__, transition, write_report, _archive_transition)
trading/position_protection.py	项目文件	类 ProtectionRepairRecord; 类 RepairPreviewProposal; 类 StopConfig; 类 RepairConfig; 类 TestThenBulkSelection; run_position_protection:运行流程; _repair_record_for_row:内部辅助; calculate_uncovered_qty:模块辅助逻辑; calculate_raw_stop_price:模块辅助逻辑; load_stop_config:读取/加载数据; load_repair_config:读取/加载数据; paper_operation_mode:模块辅助逻辑; build_repair_execution_plan:构建对象/报告; latest_test_repair_result:模块辅助逻辑; build_test_repair_result:构建对象/报告; write_test_repair_result:写出文件或结果; calculate_dynamic_stop:模块辅助逻辑; select_reference_price:筛选对象; build_repair_preview_report:构建对象/报告; select_test_then_bulk_symbol:筛选对象; build_repair_preview_proposal:构建对象/报告; write_preview_report:写出文件或结果; write_report:写出文件或结果; _blocked_reason:内部辅助; _preview_safety:内部辅助; _is_test_symbol_eligible:内部辅助; _is_run_symbol_eligible:内部辅助; _test_symbol_row:测试对应行为; _manual_test_symbol_rejection:内部辅助; `
trading/price_normalizer.py	项目文件	类 PriceNormalization; normalize_order_price:标准化数据; normalize_order_price_record:标准化数据; rounding_direction:模块辅助逻辑; log_normalization:模块辅助逻辑; _decimal_tick:内部辅助; _round_to_tick:内部辅助
trading/process_guard.py	项目文件	类 ProcessInfo(process_name); 类 ExecutionLock(__init__, heartbeat); current_execution_processes:模块辅助逻辑; stop_duplicate_autonomous_processes:停止进程/服务; write_process_guard_report:写出文件或结果; append_history:追加记录; read_lock:读取输入; process_cwd:模块辅助逻辑; process_belongs_to_project:模块辅助逻辑; _inherited_owner:内部辅助; _parent_mode9_owner:内部辅助; _pid_alive:内部辅助; _current_command:内部辅助; _utc_now:内部辅助
trading/profit_lock.py	项目文件	build_profit_lock_report:构建对象/报告; profit_lock_row:模块辅助逻辑; common_row:模块辅助逻辑; write_report:写出文件或结果; read_json:读取输入; now:模块辅助逻辑
trading/realtime_account_state_bus.py	Near real-time account-centered state bus.	类 StateUpdate; 类 RealtimeAccountStateBus(__init__, update, snapshot_payload, write_report, _archive_state_update); bootstrap_bus_from_reports:模块辅助逻辑; update_to_dict:更新状态; read_json:读取输入; utc_now:模块辅助逻辑
trading/realtime_account_sync.py	Realtime account sync orchestration for BUY/SELL synchronized decisions.	run_realtime_account_sync:运行流程; sqlite_ledger_integrity_report:模块辅助逻辑; write_report:写出文件或结果
trading/risk.py	项目文件	类 RiskLimits; 类 RiskEngine(__init__, validate, _reject)
trading/scanner_universe_builder.py	Build scanner candidates for the discovery universe.	build_scanner_universe:构建对象/报告
trading/security_master.py	项目文件	类 SecurityMasterRecord; build_security_master_report:构建对象/报告; load_seed_records:读取/加载数据; load_security_master_records:读取/加载数据; record_from_universe_row:记录数据; record_from_security_master_row:记录数据; records_from_nasdaq_trader_rows:记录数据; exchange_name:模块辅助逻辑; write_report:写出文件或结果
trading/sell_decision_engine.py	SELL decision engine. Emits SELL intents only; never calls IBKR directly.	build_sell_intent:构建对象/报告; priority:模块辅助逻辑; can_submit_order:模块辅助逻辑; submit_order:提交请求

文件	作用	关键函数/类
trading/service.py	项目文件	类 TradingService(__init__, health, audit_orders, audit_order, validate, validate_limit, _validate_proposal, submit, submit_limit, _submit_bracket, +16)
trading/session_calendar.py	项目文件	类 TradingSession; 类 SessionSnapshot; classify_trading_session:分类状态; _session_for_eastern:内部辅助; _as_utc:内部辅助
trading/session_policy.py	项目文件	类 SessionRiskPolicy; 类 SessionOrderContext; 类 PolicyDecision; default_session_policies:模块辅助逻辑; policy_for_session:模块辅助逻辑; validate_session_order:校验规则; _policy:内部辅助; _bool_env:内部辅助; _int_env:内部辅助; _float_env:内部辅助
trading/simulated_order_executor.py	Local simulated order executor.	run:运行流程
trading/simulated_pnl.py	Local simulated PnL helpers.	refresh:刷新状态
trading/simulated_position_ledger.py	Local simulated position ledger helpers.	refresh:刷新状态
trading/simulation.py	项目文件	类 StrategySimulationConfig; run_strategy_simulation:运行流程; quote_source:模块辅助逻辑; default_prices:模块辅助逻辑; default_profiles:模块辅助逻辑; value_pool_record:模块辅助逻辑
trading/sqlite_event_store.py	Append-only SQLite event store with WAL enabled.	类 SQLiteEventStore(__init__, enable_wal, journal_mode, append, append_many, _insert_with_connection)
trading/state_bootstrap_manager.py	Bootstrap buses from local cache, then keep REAL_IBKR subscriptions available.	类 StateBootstrapManager(__init__, bootstrap); source_timestamp:模块辅助逻辑; write_report:写出文件或结果; read_json:读取输入
trading/state_freshness_policy.py	Freshness policy for simulation and future IBKR paper execution readiness.	类 FreshnessThresholds(from_env); evaluate_freshness:评估决策; is_stale:布尔判断; write_report:写出文件或结果
trading/state_timestamp_reconciler.py	Timestamp reconciliation for cache, simulation, and REAL_IBKR events.	类 ReconciledField; 类 TimestampReconciler(__init__, reconcile, write_report); should_accept:模块辅助逻辑; parse_time:模块辅助逻辑
trading/strategy.py	项目文件	类 CandidateProfile; 类 ValueFilterConfig; 类 FilterResult; 类 ValuePoolFilter(__init__, evaluate, score, approved_symbols); 类 MovingAverageConfig; 类 SignalDecision; 类 MovingAverageSignalEngine(__init__, update); 类 TacticalRiskConfig; 类 StrategyPlan; 类 StrategyScannerConfig; 类 StrategyScanEvent(as_dict); 类 RotatingStrategyScanner(__init__, cache, bar_history, scan_once, _record_bar); 类 TacticalLongStrategy(__init__, plan_entry); _average_available:内部辅助; _lower_is_better:内部辅助; _higher_is_better:内部辅助
trading/strategy_modules.py	项目文件	类 ConservativeTrendConfig; 类 ModulePosition; 类 ModuleDecision(is_order); 类 PortfolioRiskConfig; 类 PortfolioRiskDecision; 类 PortfolioRiskGate(__init__, evaluate); 类 ConservativeTrendModule(__init__, positions, evaluate, record_order_decision, _exit_decision, _common_block_reason, _averages); _position_market_value:内部辅助; _is_regular_session:内部辅助
trading/strategy_signals.py	Strategy signal only architecture.	类 StrategySignal; create_strategy_signal:创建对象; write_report:写出文件或结果; submit_order:提交请求
trading/subscription_health.py	Always-listening subscription state reporting.	build_subscription_health_report:构建对象/报告; default_streams:模块辅助逻辑; stream:模块辅助逻辑
trading/take_profit.py	项目文件	build_take_profit_report:构建对象/报告; take_profit_row:模块辅助逻辑; write_report:写出文件或结果; read_json:读取输入; now:模块辅助逻辑
trading/trailing_profit.py	项目文件	build_trailing_profit_report:构建对象/报告; trailing_profit_row:模块辅助逻辑; write_report:写出文件或结果; read_json:读取输入; now:模块辅助逻辑
trading/tws_paper.py	项目文件	normalize_us_stock_price:标准化数据; 类 OrderConfirmation(acknowledged_ids, pending_confirmation, details); 类 TwsConnectionStatus(as_dict); 类 TwsPaperClient(__init__, nextValidId, run_loop, managedAccounts, error, openOrder, openOrderEnd, orderStatus); 类 TwsPaperBroker(__init__, check_status, submit_bracket, submit_limit, submit_stop, submit_stop_limit, _cancel_parent_order, _fatal_order_errors, _order_errors, _is_fatal_order_error, +7)
trading/universe.py	项目文件	类 UniverseEntry; 类 UniverseSelectionConfig; 类 UniverseSelection; load_universe:读取/加载数据; select_universe:筛选对象; default_universe_symbols:模块辅助逻辑; symbols_csv:模块辅助逻辑; _entry_from_row:内部辅助; _clean_symbol_set:内部辅助; _float_or_none:内部辅助; _int_or_none:内部辅助; _bool_or_true:内部辅助; _bool_or_none:内部辅助
trading/workflow.py	项目文件	attach_workflow_step:模块辅助逻辑; workflow_step:模块辅助逻辑; workflow_error_payload:模块辅助逻辑

.codex/hooks

文件	作用	关键函数/类
.codex/hooks/secret_scan_stop.py	项目文件	respond:模块辅助逻辑; run:运行流程; main:命令入口

backtest

文件	作用	关键函数/类
backtest/.gitkeep	项目文件	无函数

运行生成目录

目录	作用	维护规则
reports/	JSON/MD/HTML 运行报告、paper evidence、agent cycle、审计摘要。	可归档；不要在交易复盘前删除最新证据。
logs/	API、agent、pool strategy 运行日志。	排错保留；不应写入 secret。
dashboard/	静态 dashboard HTML。	由脚本生成，可重新生成。
.runtime/	后台进程 pid 和本地运行状态。	进程控制用，不作为业务事实来源。
trading_audit.sqlite3	SQLite 审计库。	不要手工编辑；用 API 或脚本读取。

7. Appendix A: API 与安全文件

文件/API	作用	安全要求
LICENSE	专有版权条款。	发给用户或协作者时保留。
SECURITY_MODEL.md	信任边界、角色权限、secret handling。	任何扩大权限前先更新。
TRADING_LOCK.md	交易锁和人工确认语义。	不要绕过 STOP/DEV/TRADE_LOCK。
AGENTS.md	Codex 工作约束和 secret scan 要求。	提交前跑 gitleaks。
api_service.py	HTTP API handler。	所有请求必须带 X-API-Key。
GET /health	API/TWS/lock/token/account readiness。	只读状态，可用于前置检查。
POST /v1/orders/validate/limit	校验 limit order。	不进 TWS，不代表成交。
POST /v1/orders/stage/limit	创建 TWS untransmitted order。	危险；可能需要手动取消。
POST /v1/orders/paper/limit	创建 transmitted paper limit order。	必须 TRADE_LOCK、token、TWS ready、风控通过。
GET /v1/orders/audit	读取审计记录。	不要直接改 SQLite。
.secrets/	本地 secret 文件。	必须 gitignore；不能提交真实凭证。
config/autonomous_paper.env.example	自动 paper 环境变量示例。	只放占位和默认值。

8. Appendix B: 术语表

术语	简明解释
API key	调用本地 Trading API 的认证密钥。
Audit ledger	记录请求、拒绝、提交、TWS 回执和耗时的审计账本。
Bar	按时间聚合的一根行情 K 线/价格条。
Bracket order	父限价单加止止损单的一组保护性订单。
Candidate	AI、扫描器或规则发现的潜在标的。
Cooldown	同一标的下单后的冷却步数/时间，避免重复触发。
Daily trade token	当天 TRADE_LOCK 生成的短期交易令牌。
DEV_LOCK	开发锁，允许检查和验证，禁止 paper transmit。
Execution gate	执行前的最终安全门，包括 token、TWS、风控、审计。
Fast SMA / Slow SMA	快/慢简单移动平均线，用于趋势确认。
IBKR	Interactive Brokers，券商接口来源。
Idempotency key	幂等键，避免同一请求重复提交。
Limit order	限定最高买入价或最低卖出价的订单。
Live trading	真实资金交易；本项目当前没有安全启用路径。
Market data type 1/3/4	IBKR 行情类型：1 实时，3 延迟，4 延迟冻结。
MODE9_RUNTIME	推荐自动运行入口，确保 TRADE_LOCK API 后启动 autonomous agent。
NBBO	美国多交易所最佳买卖报价汇总。
OpenClaw	AI 分析/客户端层，可提 proposal，不直连 IBKR。
Paper trading	模拟账户交易，不使用真实资金。
Proposal	结构化交易意图，提交给 Trading API 审核。
Quote	统一行情对象，包含 last/bid/ask/close/volume/timestamp/source。
Quote freshness	行情新鲜度；过期行情不能用于下单。
Read-only adapter	只读取行情/账户/订单状态，不提交订单。
RTH	Regular Trading Hours，美股常规交易时段。
Security master	证券主数据表，记录 symbol、交易所、类型等。
Slippage	预期价格和实际成交价格的差异。
Spread	ask 与 bid 的差额。
Stage order	已进入 TWS 但 transmit=False 的未传输订单。
STOP	最安全状态，停止 API/agent/策略并清 token。
TRADE_LOCK	允许 paper transmit 的前置锁状态。
Transmit	TWS 是否真正提交订单。
TWS	Trader Workstation，IBKR 本地交易终端。
Universe	策略可观察的股票池。
Validate	只校验请求，不进 TWS。
Value pool	按估值、质量、成长、安全性筛出的候选池。

维护规则

- 新增源文件后，同步更新本手册或重新生成第 6 节索引。
- 新增交易路径时，先更新 SECURITY_MODEL.md、TRADING_LOCK.md 和 Appendix A。
- 新增策略时，按“独立模块 -> 单元测试 -> simulated quote -> validate -> paper”的顺序推进。
- 提交或对外发送前运行 pre-commit run gitleaks --all-files；任何 secret finding 都是阻塞项。